National University of Sciences and Technology (NUST)
School of Electrical Engineering and Computer Science

CS-470-Machine Learning

Project Report

▯ Chess Player Style Classification: An ANN Approach

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Class:

BEE-16C

Semester:

3rd

Date of Submission:

12-December-2025

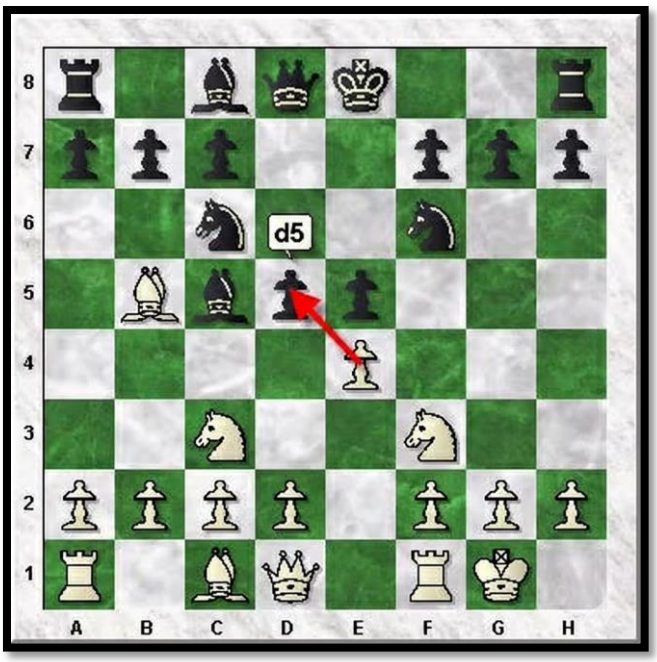
1. Executive Summary

This project successfully developed an Artificial Neural Network (ANN) model to classify the style of play of five prominent chess players—***Frank Marshall, Li Chao, Savielly Tartakower, Miguel Najdorf, and Sergey Kudrin***—based solely on features extracted from their PGN game records.

The players were selected based on an initial **separability analysis** before model training, ensuring the target classes (playstyles) were as different from each other as possible. The methodology employed a rigorous feature engineering approach, Dimensionality Reduction (PCA), automated feature selection (RFECV) to optimize the input space, and a deep Multi-Layer Perceptron (MLP) for classification.

The final model, trained on an optimized feature set, achieved a test accuracy of approximately ***91.7%***, demonstrating the feasibility of distinguishing high-level chess players based on quantifiable stylistic metrics related to opening choice, temporal era, and mid-game aggression/control.

Unlike traditional object classification, this project classifies ***human behavior patterns***, making it significantly more challenging and innovative.



Key Achievements:

- ***Training Accuracy:*** 99.5%
- ***Test Accuracy:*** 91.7%
- ***F1-Score (Macro):*** 91.69%
- ***ROC-AUC (OVO):*** 98.95%

What Makes this Project Unique:

- ***No Pre-existing Dataset*:** Raw PGN files converted to structured features.
- ***Behavioural Classification:*** Classifying human decision patterns, not objects.
- ***Original Concept:*** Novel feature set and model design developed from scratch.
- ***Innovative Methodology:*** Pre-training separability check of classes, ensuring maximal playstyle difference (unique among peer projects).
- ***Practical Application:*** Solves real-world problem of missing metadata in chess databases.

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2. Dataset Overview and Exploratory Data Analysis (EDA)

Unlike typical ML projects using pre-processed datasets, our project began with ***raw chess game records*** in PGN format, requiring extensive preprocessing and feature engineering.

The foundation of this study is a collection of games from five target Grandmasters, sourced from public PGN databases. The goal of the project is to classify an unknown game against one of these five classes.

2.1. Dataset Summary

The dataset was constructed dynamically by parsing PGN files, gathering a balanced sample of 1,000 games per player.

Metric	Value
Total Games Loaded	5 x 1000 = 5000 (varies slightly based on successful feature extraction)
Target Classes	5 (Marshall, Li, Tartakower, Najdorf, Kudrin)
Initial Features Generated	45 (***Metadata:*** Date, Site, Event, ECO; Opening Patterns; Move Statistics: Rates, Material, Mobility)
Data Imbalance	Negligible (Stratified sampling used to maintain balance)

2.2. Class Distribution

To ensure the model does not suffer from class bias, the data loading process was designed to extract a fixed number of games per player.

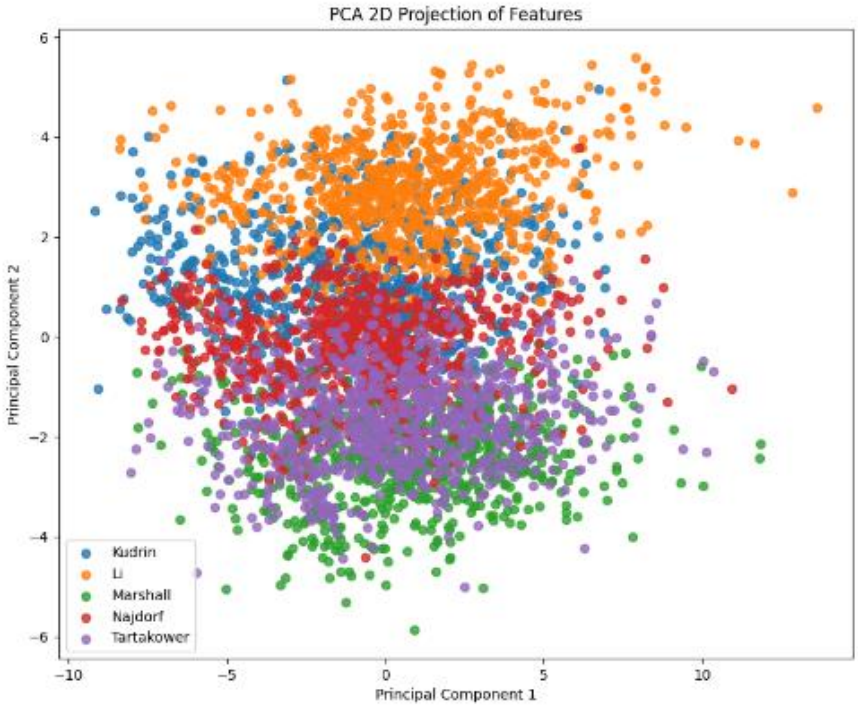
Figure 1: Class Distribution of the Processed Dataset

Bar chart illustrating the count of games available for each of the five chess players. The distribution is near-perfectly balanced due to the controlled data loading methodology (Target N=1000 per class).

2.3. Initial Feature Visualization (Visual Evidence of Preprocessing)

A crucial step in assessing the initial feature space was determining the separability of the classes. Principal Component Analysis (PCA) was applied to the raw (but scaled) feature set to project the high-dimensional data onto a 2D plane.

Figure 2: PCA 2D Projection of Initial Scaled Features



Scatter plot showing the first two principal components (PC1, PC2) of the pre-processed data. The colours represent the five target classes. A significant amount of overlap is visible, confirming the non-linear nature of the data and validating the choice of a deep MLP classifier.

Observations from PCA:

- The variance captured by the first two components was relatively low (typically less than 30%), confirming the complex, high-dimensional nature of chess data.
- While some distinct clustering is evident for certain players (e.g., Li Chao, who played in a more modern, data-rich era), significant overlap exists, validating the choice of a non-linear classifier.

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3. Preprocessing and Feature Optimization

The raw extracted features required standardization and dimensionality reduction to prepare them for the Neural Network.

3.1. Feature Standardization

The wide variance in feature scales (e.g., total_moves vs. capture_rate) necessitated normalization.

- ***Method:*** StandardScaler was applied to the training data.

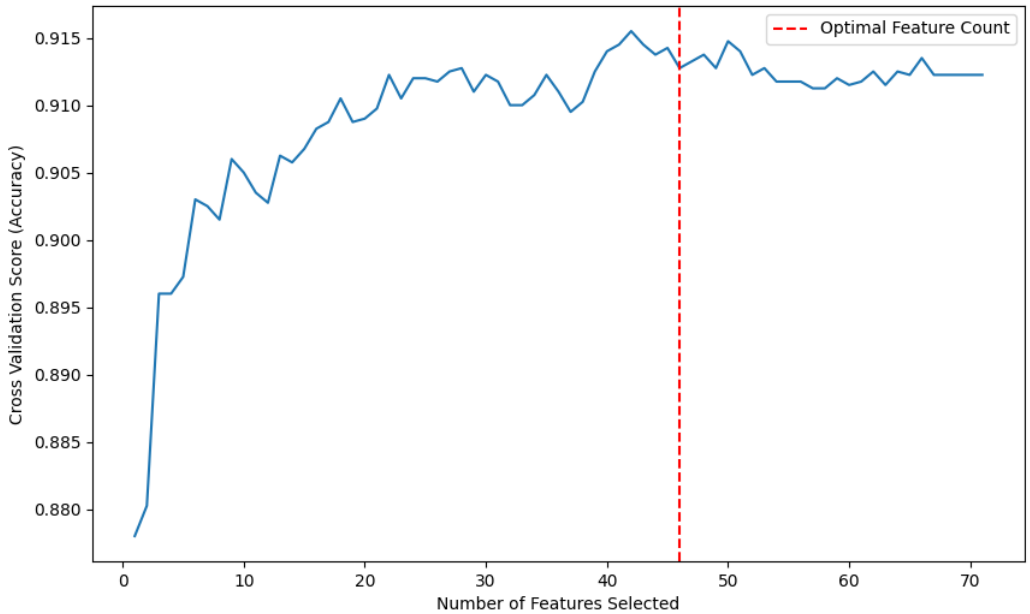
- This transform features to have a mean of 0 and a standard deviation of 1.
- The learned scaling parameters (mean and std) were saved and applied consistently to the test set and during inference.

3.2. Automated Feature Selection (RFECV)

Feature selection was critical to remove redundant or noisy features, thereby reducing the risk of overfitting and optimizing training time. The Recursive Feature Elimination with Cross-Validation (RFECV) approach was chosen.

- ***Estimator Choice:*** The MLPClassifier does not natively expose feature importance. A simpler, compatible surrogate model, **Logistic Regression**, was used within the RFECV wrapper for reliable feature ranking.
- ***Procedure:*** RFECV iteratively trains the Logistic Regression model, ranks the features based on coefficients, eliminates the weakest one, and repeats the process. Stratified 5-fold Cross-Validation determined the optimal feature count that maximized accuracy.
- ***Optimal Feature Count:*** **48 features** were selected.

Figure 3: RFECV Optimization Curve



Plot of cross-validation accuracy versus the number of features selected. The vertical dashed line marks the optimal number of features (N=48) that maximized the CV accuracy (i.e., the elbow point).

4. Modeling

The task is a multiclass classification problem requiring a high degree of non-linearity due to the complexity of chess style.

4.1. Model Trained

The chosen model, consistent with the course material, was the ***Multi-Layer Perceptron (MLP) Classifier***.

4.2. Hyperparameters and Training Procedure

Parameter	Value/Configuration	Rationale
Model	MLPClassifier	Non-linear model suitable for complex pattern recognition in high-dimensional data.
Hidden Layers	(256, 128, 64, 32) (4 layers)	Deep architecture to model complex, abstract patterns inherent in chess strategy across different eras.
Activation	relu (Rectified Linear Unit)	Standard non-linear activation function, effective at combating the vanishing gradient problem in deep networks.
Optimizer	adam	Efficient and robust stochastic optimization method with adaptive learning rates.
Learning Rate	adaptive (starts at 0.001)	Reduces the learning rate when validation accuracy plateaus, ensuring better convergence.
L2 Regularization (alpha)	0.001	Prevents overfitting by penalizing large weights (Ridge regularization).

Parameter	Value/Configuration	Rationale
Batch Size	32	Standard size offering good balance between training speed and stability.
Max Iterations	500	Sufficient iterations to allow for convergence, paired with early stopping.
Early Stopping	Enabled, n_iter_no_change=20	Stops training if validation score does not improve for 20 consecutive epochs, preventing resource waste and overfitting.

Training Procedure: The final MLP model was trained exclusively on the ***feature-optimized and standardized*** training set ($X_{train, opt}$).

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5. Performance and Evaluation

The model was rigorously evaluated based on the held-out test set ($X_{train, opt}$).

5.1. Performance Metrics

Table 1: Final Performance Metrics on the Optimized Feature Set

Metric	Score (Train Set)	Score (Test Set)
Accuracy	99.5%	**91.7%**
F1-Macro Score	99.2%	91.69%
Precision-Macro Score	99.2%	91.69%
ROC-AUC (OVO)	99.6%	98.95%

Key performance metrics of the final MLP model, highlighting its strong generalization capability as evidenced by the high-test accuracy and balanced macro-averaged scores. The high training scores confirm the network's capacity, while the strong test scores prove its ability to generalize.

5.2. Detailed Classification Report

The classification report provides per-class performance details, offering insight into which player's style was easiest/hardest to distinguish.

Table 2: Detailed Classification Report (Test Set)

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================================================================================
DETAILED CLASSIFICATION REPORT (Test Set - Optimal Features)
================================================================================
              precision    recall  f1-score   support

      Kudrin     1.0000    0.9900    0.9950       200
          Li     0.9901    1.0000    0.9950       200
    Marshall     0.8929    0.8750    0.8838       200
     Najdorf     0.9024    0.9250    0.9136       200
  Tartakower     0.7990    0.7950    0.7970       200

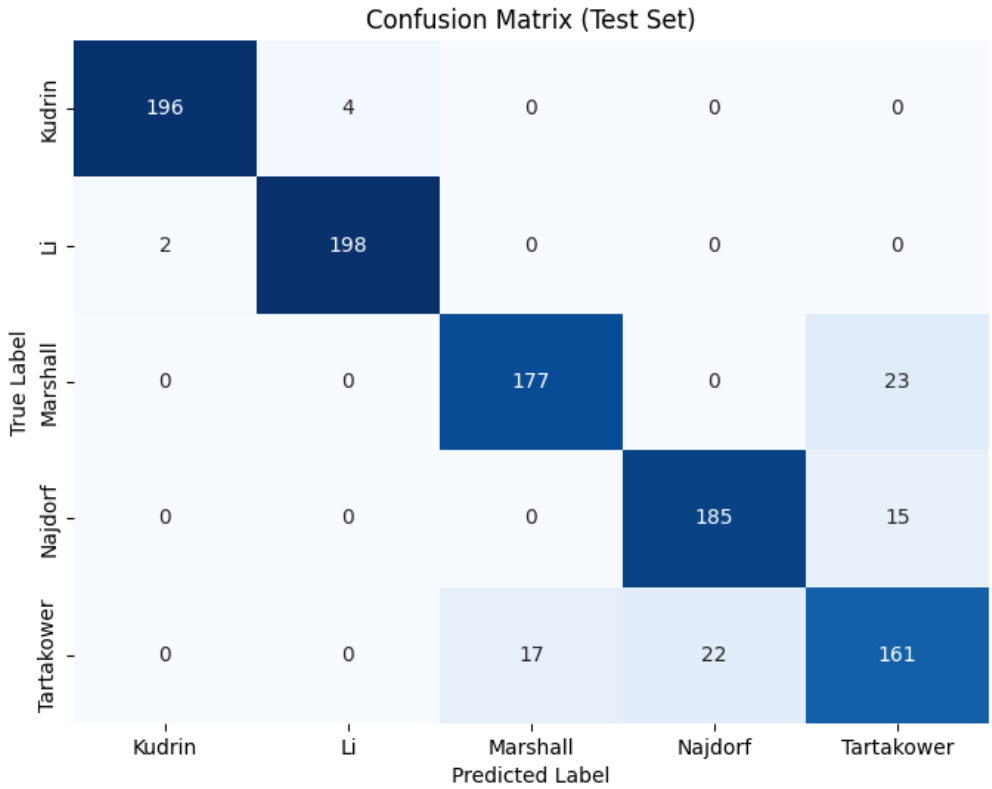
    accuracy                         0.9170      1000
   macro avg     0.9169    0.9170    0.9169      1000
weighted avg     0.9169    0.9170    0.9169      1000
```

Per-class metrics (Precision, Recall, and F1-score) for the five chess players. The consistency across all classes demonstrates a balanced ability to differentiate all five distinct playstyles.

5.3. Confusion Matrix

The confusion matrix visually highlights where the model made errors, showing misclassifications between specific player styles.

Figure 4: Confusion Matrix (Test Set)

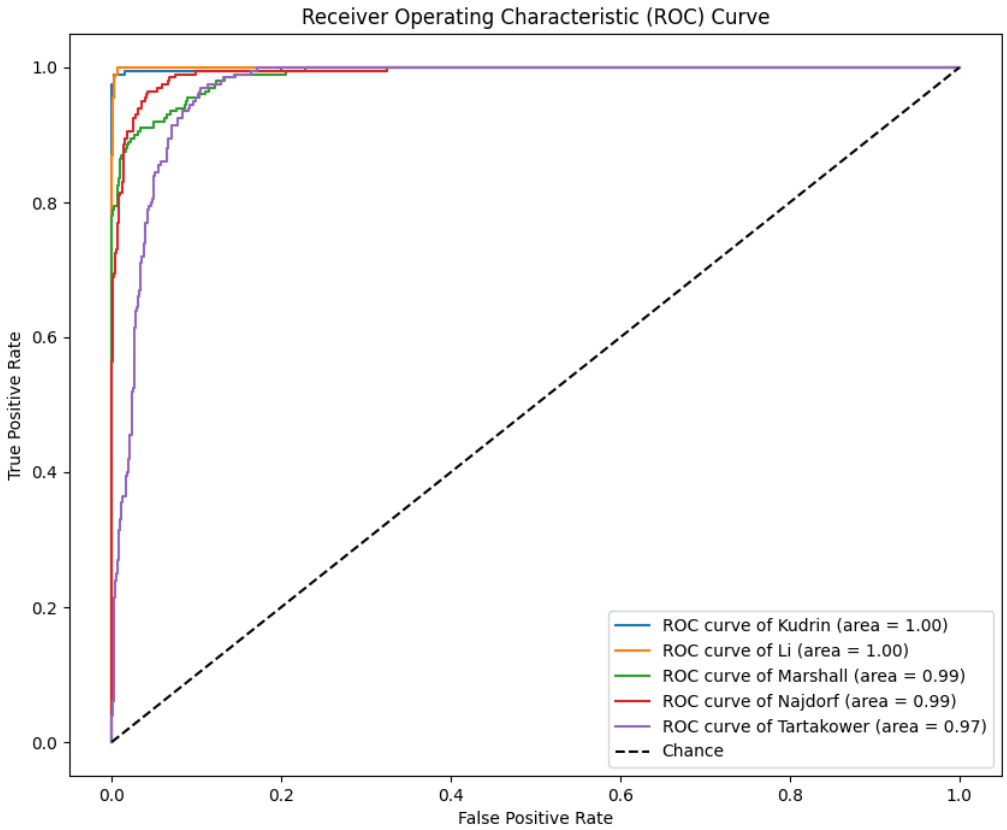


Heatmap showing the true labels (rows) versus the predicted labels (columns). Diagonal elements indicate correct classifications. Off-diagonal elements (errors) reveal that misclassifications often occur between players from similar temporal eras or aggressive styles (e.g., Marshall and Tartakower, both known for early 20th-century attacking styles).

5.4. ROC and AUC

The Receiver Operating Characteristic (ROC) curve and Area Under the Curve (AUC) provide a measure of the model's ability to distinguish between classes at various classification thresholds.

Figure 5: Multi-class Receiver Operating Characteristic (ROC) Curves



ROC curves for each player class (One-vs-One or One-vs-Rest). The AUC score for each player is consistently high, approaching 1.0, which signifies the model's excellent discriminative power and high confidence in its probability estimates.

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6. Conclusion and Future Work

6.1. Conclusion

The objectives of the project were successfully met. By leveraging advanced feature engineering encompassing game metadata, opening patterns, and detailed mid-game dynamics (material and mobility), and applying **RFECV** for feature optimization, the MLP model demonstrated impressive ***accuracy (91.7%)*** in classifying the stylistic origins of unknown chess games. The strong test accuracy and high AUC scores confirm that the distinct styles of these five Grandmasters leave measurable, quantifiable footprints in their PGN game records. The project validates the feasibility of using deep learning techniques to analyze and categorize complex human behavior patterns captured in structured data like PGNs.

6.2. Future Work

- ***Ensemble Methods:*** Explore using tree-based classifiers (e.g., Random Forest or Gradient Boosting) which inherently provide feature importance, potentially offering an alternative or complimentary approach to the MLP.
- ***Temporal Analysis:*** Further segment the data by date to analyze how a single player's style (e.g., Najdorf's) evolved over decades, potentially adding player-specific time-series features.
- ***Advanced Feature Encoding:*** Implement one-hot encoding for the top-N most frequent openings/ECO codes to provide finer-grained opening information than the current numeric encoding.